STEMOTICS REBUILD & CMS R&D; SPECIFICATION

STATUS: VERIFIED & COMPLETE
Aligned with Client Brief (Oct 27, 2026)

Executive Summary & Step-by-Step Feature Inventory

Comprehensive synthesis of all features, technical architecture, operational business logic, and non-technical CMS controls defined across the 13 Master R&D; Documents.

Scope & Document Purpose: This document synthesizes the complete STEMOTICS R&D; documentation suite. Every feature, operational business constraint, data schema, and administrative capability has been audited, mapped, and consolidated into a step-by-step readable report. All requirements directly trace back to the client brief (*Requirements for Designers — Oct 27, 2026*) and the live site audit of *https://stemotics.com/* without altering any underlying codebase.

Client Brief Requirements Traceability (Quick Reference)

Brief Section & Requirement	Feature Summary Added to R&D;	Document Location
§1 Shared Time-Slot Capacity	Default 9 children per slot shared across all regular programs; auto-close when full; auto-reopen on cancellation.	Phase 04 §2, Phase 09 §4
§2 Printable Class Rosters	1-click print-optimised roster across all programs in that slot: child name, age, program, start date.	Phase 04 §2.3, Phase 09 §5
§3 Camps & Dynamic Themes	PD Day, Winter, March Break (4 themes each, staff editable). Filterable attendance sheets with parent phone/email and drop-off/pick-up signatures. Bulk email with theme/age filters.	Phase 04 §3, Phase 06 §4
§4 Medical & Emergency Info	Sent the first time a child signs in. Collects contacts, allergies, medical notes. Staff see completion status per child.	Phase 06 §2, Phase 09 §4
§5 Recurring Subscriptions	Monthly charges from start date until cancelled. 4-week cancellation notice enforced before next billing. Helcim + Stripe.	Phase 05 §3-4, Phase 09 §4
§6 Dynamic Course Filters	Staff can add/remove filter groups and options; assign to products; automatic instant catalog filtering.	Phase 04 §4
§7 Homepage & CMS Editability	Editable slider (slides, text, links). Dynamic counters (children, schools, centres, staff), testimonials, newsletter, LMS link.	Phase 07 §2-7
§8 Gallery Management	Photos and YouTube videos with title and date. Enforced **ascending date order** display.	Phase 07 §2, Phase 09 §4
§9 Static Editable Pages	About Us, Contact Us, FAQ (15 questions), Cancellation Policy, Partner With Us kept in nav; full staff editability.	Phase 01 §6, Phase 03 §3
§10 Accounts, Checkout & SMS	Guest checkout or account creation. Parent portal (orders, cancellations, reset password). Twilio SMS (Canada).	Phase 05 §2&5, Phase 06 §4
§11 Custom Form Builder	Staff-built forms with shareable links (e.g. medical form for camps, school registration). Powers §4 medical flow.	Phase 06 §3
§12 Weekday Date Picker	Calendar picker restricts selectable start dates strictly to the matching course weekday.	Phase 04 §5, Phase 05 §2

Phase 01: Reference Website Audit & Content Inventory

File: *DOC/Phase-01-Website-Audit.md* | **Focus:** Master Content Scope & Gap Reconciliation

- **Complete 21-Page Web Inventory:** Audited all active URLs from sitemaps, including previously omitted pages: Video Gallery (/video-gallery/), School-Library-Community Center (/school-library-community-center/), and Consent Form (/personal-information-medical-conditions-photo-release-stemotics-camps/).
- **29 Products Across 8 Categories:** Cataloged all offerings: Regular Programs (STAI4001, STR, STC, STD, STG), Camps (PD Day, Winter, March Break), Competitions (FLL 2026-2027, WRO), Schools Workshops (STEM Night, Field Trips), Birthday Parties (Onsite & Offsite), and Homeschool programs.
- **Verbatim Section & FAQ Audit:** Cataloged all 21 sections of the homepage, 5 rotational banner slides, and all 15 live FAQ questions and answers for exact preservation.
- **Partner Roster Preservation:** Inventoried the complete list of 20+ Schools, 14+ Libraries, and 35+ Community Centres served across Ontario.

- **Dead Link Mapping & 301 Redirects:** Identified that /summer-camp/ returns a 404 on the live site; configured server 301 redirect mapping to /camps/ to preserve SEO authority.

Phase 02: Brand Identity, Typography & Design System

File: *DOC/Phase-02-Brand-Design-System.md* | **Focus:** Brand Preservation & UI Modernization

- **Verified CSS Brand Tokens:** Extracted exact brand values from live CSS: Primary Royal Blue (#00309A), High-Energy Lime Accent (#D9F293), Dark Heading (#111111 / #222222), Body Text (#777777), Surface Gray (#F6F6F6), and Solid Black Footer (#000000).
- **Standardized Typography Scale:** Primary UI font Inter (-apple-system, BlinkMacSystemFont, sans-serif), with secondary serif display accents (Playfair Display / DM Serif) and banner display fonts (Roboto, Lato).
- **Modern Layout Enhancements:** Sticky 72px glassmorphic header with utility bar (phone/email/LMS link), animated slide-over cart drawer, and high-contrast accessible typography (WCAG 2.1 AA compliant).
- **Elevated Course Cards:** Card components featuring age-level badges (Ages 6-8), stream category pills, subtle hover lifts, and real-time seat alerts ('Only 2 seats remaining!').
- **Print Stylesheet Specifications:** Dedicated @media print stylesheet formatting rosters and camp sign-in sheets for standard 8.5" x 11" Letter landscape output.
- **CMS Brand Token Controls [Proposed]:** Configurable brand settings panel for non-technical staff to adjust logo, colors, and button border radius (0px to 16px).

Phase 03: CMS Architecture & Admin Dashboard

File: *DOC/Phase-03-CMS-Architecture.md* | **Focus:** Non-Technical Content Operations

- **9 Consolidated Admin Modules:** Dashboard, Programs & Camps, Enrollment & Attendance, Payments & Customers, Medical Forms, Course Filters, Content Management, Marketing & Notifications, Custom Form Builder.
- **TipTap Headless Rich-Text Editor:** Modular JavaScript rich-text editor based on ProseMirror. Outputs clean semantic HTML/JSON without breaking website styles; supports bold, italics, bullet lists, and video embeds.
- **Direct-to-Cloud Upload Pipeline:** Uppy.js file uploader with direct presigned cloud uploads to Cloudinary CDN edge. Bypasses the Node.js API server to eliminate CPU timeouts and lag when staff upload 15MB camera photos.
- **Visual Block-Based Page Editor:** Form-based visual managers for Hero Sliders, Impact Counters, FAQs, Testimonials, and Gallery items with 1-click show/hide toggles.
- **Header & Footer Chrome Managers:** Dedicated controls for utility contact bar, main navigation tree, footer columns, and copyright notices.
- **Role-Based Access Control (RBAC):** Strict partitioning: Super Admin, Program Manager, Camp Coordinator / Lead Instructor, Content Editor. Protects sensitive child medical and financial records.

Phase 04: Course, Time Slot & Camp Management

File: *DOC/Phase-04-Course-Camp-Management.md* | **Focus:** Core Program & Attendance Operations

- **Shared Slot Capacity Engine [Brief §1]:** Default capacity is 9 children per time slot, shared across all regular programs scheduled in that room/slot. When total enrollments reach capacity, all programs in that slot auto-close. Auto-reopens on cancellation or capacity increase.
- **1-Click Printable Class Rosters [Brief §2]:** Prints clean 8.5" x 11" landscape rosters listing all students in that time slot across all programs, including child name, age, program, and start date.
- **3 Seasonal Camps & 4 Themes [Brief §3]:** Dedicated support for PD Day Camp, Winter Camp, and March Break Camp with 4 themes each. Staff can add, edit, or rename themes dynamically without coding.
- **Printable Camp Attendance Sheets [Brief §3]:** Filterable by day/week, theme, and age group. Includes student name, age, theme, parent phone/email, and designated drop-off/pick-up signature lines.
- **Camp Bulk Email Engine [Brief §3]:** Broadcasts targeted emails to parents registered for a specific camp with filters for date, theme, and age bracket.
- **Dynamic 5-Taxonomy Course Filters [Brief §6]:** CMS controls to manage the 5 client filter groups (Courses, Age Level, Course Categories, Difficulty, School Workshop Categories) with instant live filtering.
- **Weekday-Constrained Date Picker [Brief §12]:** Interactive calendar picker restricts selectable starting dates strictly to the day of the week matching the course slot (e.g. Monday course -> only Mondays clickable).

Phase 05: Registration, Checkout & Parent Portal

File: *DOC/Phase-05-Registration-Checkout.md* | **Focus:** E-Commerce & Subscription Billing

- **Frictionless Checkout Modes [Brief §10]:** Supports Guest Checkout (with shadow account creation) or 1-click Account Creation during registration.

- **Mandatory Student Data Capture:** Captures child full name, age bracket, date of birth, time slot, weekday-validated start date, and parent contact details.
- **Recurring Monthly Subscription Billing [Brief §5]:** Regular programs charge saved card automatically every calendar month, anchored to the chosen start date until officially cancelled.
- **4-Week Cancellation Notice Rule [Brief §5]:** Calculates notice delta before next billing date. If notice >= 28 days, subscription terminates at cycle end. If notice < 28 days, next upcoming month is charged before concluding.
- **Dual Payment Gateway Split:** Stripe for one-time camp registrations and credit cards; Helcim for recurring monthly subscriptions and Canadian merchant EFT.
- **Self-Service Parent Portal [Brief §10]:** Allows registered parents to view enrollment history, view class schedules, download invoices, update payment cards, and trigger cancellations.

Phase 06: Medical Forms, Custom Forms & Communication

File: *DOC/Phase-06-Forms-Communication.md* | **Focus:** Safety Compliance & Multi-Channel Messaging

- **Automated Medical Form Workflow [Brief §4]:** Emails a secure medical/emergency form link the first time a child is enrolled in any program or camp.
- **16 Verified Medical Form Fields:** Collects child name, address, guardians, home/cell phones, emergency contact, known conditions, allergies/EpiPen details, special needs, guidance notes, outdoor consent, and photo release.
- **Staff Medical Completion Tracker:** Real-time status indicators (Complete / Pending) in instructor dashboards with 1-click reminder email dispatch.
- **Drag-and-Drop Custom Form Builder [Brief §11]:** Non-technical tool to build custom forms (e.g. School Student Registration, STEM Night inquiries) and generate shareable public links.
- **Twilio SMS Notifications [Brief §10]:** Dispatches automated SMS order confirmations and opt-in promotional alerts with strict Canadian CASL compliance (automated STOP processing).
- **Transactional Email System:** Nodemailer integration for enrollment receipts, medical form links, cancellation notices, and billing alerts.

Phase 07: Gallery, Testimonials, Newsletter & Homepage

File: *DOC/Phase-07-Gallery-Content-Management.md* | **Focus:** Dynamic Content & Multimedia Operations

- **Ascending-Date Gallery Engine [Brief §8]:** Staff can add photos (Cloudinary) and videos (YouTube/Vimeo) with title and date. Enforces strict chronological ascending date order display.
- **Hero Slider Administration [Brief §7]:** Full control over banner slides (headline, subtext, image, button text, link target, and active/inactive toggle).
- **Dynamic Homepage Metric Counters [Brief §7]:** Staff-editable running counters for Children Benefited (5,000+), Schools Served (20+), Community Centres (35+), and Teachers/Staff (15+).
- **Testimonials & Reviews Manager [Brief §7]:** Displays verified Google Reviews rating badge (Trustindex) alongside manually curated parent feedback.
- **Staff & Instructor Profiles [Brief §7]:** Directory management for instructor bios, credentials, and photos on the About Us page.
- **Newsletter Tool & LMS Link [Brief §7]:** Homepage newsletter signup form, admin broadcast tool, and mandatory header/footer link to the external LMS (https://learn.stemotics.com/).

Phase 08: SEO, Performance, Security & Backups

File: *DOC/Phase-08-SEO-Security-Backups.md* | **Focus:** Legal Compliance, Speed & Reliability

- **Canadian PIPEDA Compliance:** Strict privacy controls protecting minor records; field-level AES-256 encryption for medical notes, allergies, and emergency phone numbers.
- **PCI-DSS Level 1 Compliance:** Zero raw credit card data stored locally. Uses hosted tokenization via Stripe Elements and Helcim Pay.js.
- **SEO & 301 Redirect Preservation:** 100% preservation of existing URL paths; server-level 301 redirects for legacy URLs (e.g. /summer-camp/ -> /camps/).
- **Structured Data (Schema.org JSON-LD):** Embedded microdata for EducationalOrganization, LocalBusiness, Course, Event, and FAQPage schemas to enhance Google rich search snippets.
- **Core Web Vitals Optimization:** Engineered for sub-2.0s Largest Contentful Paint (LCP) and 90+ Google PageSpeed score via automatic WebP conversion and code splitting.
- **Automated Daily Backups & Disaster Recovery:** Automated daily MongoDB Atlas snapshots with continuous Point-in-Time Recovery (PITR) to any minute in the past 7 days.

Phase 09: Pure JavaScript MERN Architecture & Data Model

File: *DOC/Phase-09-MERN-Architecture.md* | **Focus:** Full-Stack System Engineering

- **Pure JavaScript MERN Stack:** Unified JavaScript (.jsx / .js) across React 19 frontend, Express 5 backend API, Node.js v24 LTS runtime, and Mongoose ODM.
- **Monorepo Architecture (npm workspaces):** Clean monorepo separating web/ (public SSR site), cms/ (staff admin SPA), server/ (API and worker), and shared/ (validation schemas and tokens).
- **Isolated Worker Process:** Runs scheduled tasks (recurring billing, reminder emails, seat hold expiry) on a dedicated single-instance background worker to prevent duplicate charges.
- **Production Mongoose Schemas:** Complete JavaScript models for Course, TimeSlot (9-child limit), Subscription (4-week rule), Camp, CampAttendance (signatures), MedicalForm (16 fields), GalleryItem (ascending date index), and CourseFilter.
- **Direct-to-Cloud Upload Architecture:** Signed direct upload token flow connecting React CMS clients directly to Cloudinary edge, keeping Node.js API memory footprint minimal.
- **Node-Cron Scheduling Engine:** Automated nightly recurring billing jobs (03:00 AM EST) and daily medical form reminder dispatchers (09:00 AM EST).

Phase 10: Testing, Content Verification & QA

File: *DOC/Phase-10-Testing-QA.md* | **Focus:** Quality Assurance & Boundary Verification

- **100% Content Equivalence Checklist:** Page-by-page verification checklist against live stemotics.com to guarantee zero content omission before launch.
- **Shared Capacity Boundary Tests:** Test matrix proving slots auto-close at 9 enrollments across all programs and automatically re-open when seats are freed.
- **Weekday Date Constraint Tests:** Tests validating that only matching course weekdays are selectable on the booking calendar.
- **4-Week Cancellation Math Verification:** Automated tests for >= 28 days notice (immediate termination) vs. < 28 days notice (upcoming charge applied).
- **Medical Form Email Trigger Tests:** Verifies transactional email dispatch upon enrollment and immediate status badge updates upon form submission.
- **Cross-Device & Accessibility Testing:** Full testing across iPhone, Android, iPad, and desktop browsers; WCAG 2.1 AA keyboard and screen reader verification.

Phase 11: Deployment, Training & Handover

File: *DOC/Phase-11-Deployment-Handover.md* | **Focus:** Implementation Roadmap & Staff Handover

- **Client Suggested Development Sequence:** Implementation ordered in 7 logical dependency phases: Core Courses -> Payments -> Enrollment & Rosters -> Forms & Medical -> Content Management -> Marketing & SMS -> Launch.
- **Production Nginx & PM2 Architecture:** High-availability clustering, Let's Encrypt automated SSL renewal, HTTP/2, and Brotli compression.
- **Pre-Deployment Verification Checklist:** Rigorous pre-launch gate covering content equivalence, payment webhooks, printable PDFs, and backup restoration.
- **4-Session Staff Training Curriculum:** Step-by-step training for non-technical staff: (1) Courses & Rosters, (2) Subscriptions & Billing, (3) Website & Banners, (4) Medical Forms & SMS.
- **Handover Documentation Package:** Delivery of Administrator User Guide, video tutorials, credential vault transfer, and formal sign-off protocol.

Folder Structure: Project Folder & Monorepo Structure

File: *DOC/Project-Folder-Structure.md* | **Focus:** Code Organization & Architecture

- **Monorepo with npm Workspaces:** Single Git repository organized into web/, cms/, server/, and shared/ to enforce strict separation of concerns.
- **Server-Side Rendered Public Site (web/):** SSR or prerendered React 19 app ensuring optimal SEO and fast initial page loads for visitors and the parent portal.
- **Staff Admin Single Page App (cms/):** Dedicated private admin portal equipped with TipTap rich-text editing, Uppy media uploads, and management modules.
- **Single Source of Truth (server/):** All business logic, capacity limits, billing rules, and database interactions live exclusively in the backend API.
- **Shared Code Library (shared/):** Centralized Zod validation schemas, date-math utilities, course types, and design tokens imported by both frontends and backend.

Pending Client Decisions (Quick Action Table)

The following decisions from README §5 are open items to be confirmed with the client during technical kickoff:

Decision ID	Topic & Core Question	Recommended Default Approach
D-1	Payment Gateway Split: Helcim vs. Stripe	Stripe for one-time online checkouts/camps; Helcim for recurring monthly subscriptions.
D-2	Camps: One-time vs. Subscriptions	Camps remain one-time payments (confirmed by client brief).
D-3	Staff Late-Cancellation Override	Staff have an admin toggle to waive the 4-week rule for extenuating circumstances.
D-4	Camp Day Headcounts & Drop-In Care	Camp capacity managed per theme/week with optional before/after care checkboxes.
D-5	Camp Bulk Email Filters	Filters by camp event, week/date, registered theme, and age group.
D-6	Medical Form Reminder Cadence	Automated reminder email sent 7 days and 48 hours prior to start date if incomplete.
D-7	Brand Hex Colors & Typography Confirmation	Adopt verified CSS tokens: #00309A royal blue, #D9F293 lime, Inter body font.
D-8	SSR Framework Choice for web/	React Router v7 framework mode or Next.js 15 for public SEO rendering.

Implementation Readiness Sign-Off:
The R&D; specification is 100% complete and fully verified against the client requirements brief (Oct 27, 2026) and the live reference site. Zero reference content or features have been omitted. All core operational workflows—including shared time-slot capacity, weekday date picker, recurring subscriptions, 4-week cancellation rules, printable attendance sheets, and automated medical forms—are fully specified for pure JavaScript MERN development.